

WITH anchor\_candidate AS (

```
SELECT
  party_id,
  first_alert_date,

  LAST_DAY(
    ADD_MONTHS(first_alert_date, -3)
  ) AS raw_anchor_date
```

```
FROM bu_frd.party_first_valid_alert_all
),
```

anchored AS (

```
SELECT
  party_id,
  first_alert_date,

  CASE
    WHEN raw_anchor_date < DATE '2022-01-31'
      THEN DATE '2022-01-31'
    ELSE raw_anchor_date
  END AS observation_date
```

```
FROM anchor_candidate
),
```

x AS (

```
SELECT
  party_id,
  first_alert_date,
  observation_date,

  CASE
    WHEN first_alert_date <= observation_date
      THEN '01_ALERT_ON_OR_BEFORE_OBS'

    WHEN first_alert_date <=
      LAST_DAY(ADD_MONTHS(observation_date,3))
      THEN '02_VALID_POSITIVE'

    ELSE '03_AFTER_3M_WINDOW'

  END AS status
```

```
FROM anchored
```

```

WHERE observation_date BETWEEN
    DATE '2022-01-31'
    AND DATE '2025-12-31'
)

SELECT
    status,
    COUNT(*) AS row_count,
    COUNT(DISTINCT party_id) AS party_count

FROM x

GROUP BY status

ORDER BY status;

```

#####Second#####

```

DROP TABLE IF EXISTS bu_frd.model_positive_anchor_3m_2022_2025;

```

```

CREATE TABLE bu_frd.model_positive_anchor_3m_2022_2025
USING PARQUET
AS

```

```

WITH anchor_candidate AS (

```

```

    SELECT
        party_id,
        first_alert_date,

        LAST_DAY(
            ADD_MONTHS(first_alert_date,-3)
        ) AS raw_anchor_date

```

```

    FROM bu_frd.party_first_valid_alert_all
),

```

```

anchored AS (

```

```

    SELECT
        party_id,
        first_alert_date,

        CASE

```

```

        WHEN raw_anchor_date < DATE '2022-01-31'
        THEN DATE '2022-01-31'
        ELSE raw_anchor_date
    END AS observation_date

    FROM anchor_candidate
)

```

```

SELECT
    party_id,
    first_alert_date,
    observation_date,

    YEAR(observation_date)
    AS observation_year,

    CASE
        WHEN YEAR(observation_date) IN (2022,2023)
        THEN 'TRAIN'

        WHEN YEAR(observation_date) = 2024
        THEN 'VALIDATION'

        WHEN YEAR(observation_date) = 2025
        THEN 'TEST'
    END AS model_split,

    1 AS positive_flag

```

```

FROM anchored

```

```

WHERE observation_date BETWEEN
    DATE '2022-01-31'
    AND DATE '2025-12-31'

```

```

AND first_alert_date > observation_date

```

```

AND first_alert_date <=
    LAST_DAY(
        ADD_MONTHS(observation_date,3)
    );

```

```

#####third#####

```

```

SELECT
    model_split,
    observation_year,
    COUNT(*) AS rows,

```

```

COUNT(DISTINCT party_id) AS parties

FROM bu_frd.model_positive_anchor_3m_2022_2025

GROUP BY
    model_split,
    observation_year

ORDER BY observation_year;

#####four#####

DROP TABLE IF EXISTS bu_frd.model_positive_anchor_3m_2022_2025_sample01;

CREATE TABLE bu_frd.model_positive_anchor_3m_2022_2025_sample01
USING PARQUET
AS

SELECT *

FROM bu_frd.model_positive_anchor_3m_2022_2025

WHERE PMOD(
    HASH(CAST(party_id AS STRING)),
    10000
) < 100;

###validate###

SELECT
    model_split,
    observation_year,

    COUNT(*) AS positive_count,

    COUNT(DISTINCT party_id)
    AS positive_party_count

FROM bu_frd.model_positive_anchor_3m_2022_2025_sample01

GROUP BY
    model_split,
    observation_year

ORDER BY observation_year;

```

